Hacktiv8

Optimizing Book Exchange Inventory and Promotion Strategy

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Project Description

Background

A **book-exchange platform** operates by matching supply (available books) with demand (books requested for swapping) across different countries. However, the platform currently **lacks data-driven insights** to determine which books should be prioritized for inventory acquisition, promotion, and regional marketing.

With a large dataset containing book attributes such as genre, ratings, publication year, bestseller status, and movie adaptations, there is an opportunity to uncover demand patterns and optimize business decisions.

Problem Statement

The client needs a data-driven analysis to identify and compare the most popular types of books for exchange across different countries, measured through metrics such as average ratings, popularity distribution, and bestseller frequency, to optimize inventory planning and promotional strategies within two weeks for quarterly decision-making.

1. Specific

The client needs to identify which types of books (by genre, rating, age category, and adaptation status) are most popular for exchange across different countries.

2. Measurable

Success will be measured by the ability to rank and compare books and categories using metrics such as average ratings, popularity distribution, and bestseller frequency.

3. Achievable

The dataset contains sufficient variables (ratings, genres, popularity country, bestseller status) to conduct both descriptive and inferential analysis.

4. Relevant

Improving inventory planning and promotion strategies directly impacts exchange success rates, user satisfaction, and operational efficiency.

5. Time-Bound

The analysis and dashboard will be completed within two weeks and used for quarterly strategic planning.

Project Objectives

- Identify the most in-demand book characteristics for exchanges.
- Understand how popularity varies by periods of time and genre.
- Analyze whether high ratings and bestseller status significantly influence exchange demand.
- Provide actionable insights through dashboard for decision-making.

Project Usage

- Support inventory acquisition decisions.
- Guide promotional campaigns.
- Optimize regional marketing strategies.
- Improve recommendation and swapping success rates on the platform.

Dataset Description

The dataset is taken from the Kaggle Website. The dataset titled "**The Most Popular Books for Exchanging**" contains information on books that are frequently requested or swapped on a book-exchange platform. Each record represents a single book with complete descriptive and popularity indicator information.

The dataset contains **990 records** of book with **18 columns**/attributes.

Column **awards has 600 missing values** and column **movie_release_year has 350 missing values**, the rest of the columns don't have missing values.

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Analysis Results

Problems Breakdown

1. Which book genres are most popular for exchanging?

Based on the top 10 genres among 81 genres, **fantasy** is the most popular genre for book exchanges by a clear margin, followed by **sci-fi** and **historical fiction**. However, genres such as horror, dystopian, and mystery show relatively lower exchange popularity compared to the top genres.

2. Do books with higher average ratings tend to be more popular for exchanges?

Based on the chart, books with **medium ratings** (around 3.75–4.25) tend to have higher exchange counts, indicating greater popularity. However, very highly rated books have less exchange counts. In conclusion, **ratings do influence popularity** but are not the sole factor exchange demand.

3. Are books adapted into movies more popular for exchange compared to non-adapted books?

Books with movie adaptation has significantly higher number of exchanges than non-adapted books. So, **movie adaptations will increase book's popularity and demand** on the exchange platform.

Tableau Dashboard:
https://public.tableau.com/app/profile/clarence.manzo/viz/P0M1_clarence_manzo/BookExchangeInsightsDashboard?publish=yes

Problems Breakdown

4. How does the popularity of exchanged books vary by publication year?

Book exchange popularity for the very old publication years (below 1750) is very low and start to increases from around the year 1800. Than, it increases sharply for more recent years after the mid-1900s. It means that **newer and more modern books** tend to be exchanged more than older books.

5. How is the data distribution of book ratings in the exchange platform?

The book ratings average and median are the same, indicating that book ratings is **normally distributed**. The standard deviation indicates **very low variability** in ratings, meaning users generally rate books consistently high. The skewness value shows that the ratings distribution is **symmetric**. The kurtosis shows a **slight peaked distribution** as the ratings are clustered in the center with few extreme values.

6. Is there a statistically significant difference in average ratings between bestseller and non-bestseller books?

There is a **significant difference** in average ratings between bestseller and non-bestseller books. It implies that bestseller books tend to be rated differently compared to non-bestseller books.

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Conclusion

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The analysis shows that **fantasy**, **sci-fi**, and **historical fiction** should be prioritized. Books with **medium ratings** perform better than extremely high-rated books, indicating that popularity is influenced by broader accessibility and audience reach. Books with **movie adaptations** is great for promotion due to the high demand. Additionally, **newer books** should be focused more than older books

Book ratings across the platform are stable, providing a reliable basis for comparison. **Bestseller** books should be prioritized as well. These insights will improve the platform in inventory prioritization, promotional focus, and strategic planning.